

练习二十五 电磁感应与电磁波（一）

1. $\frac{1}{18}B\omega L^2$; $\frac{2}{9}B\omega L^2$; $\frac{1}{6}B\omega L^2$

2. $\frac{\mu_0 I v}{2\pi} \ln 3$; N

3. (2)

4. (1)

5. 通过矩形线圈的磁通量 $\Phi = \frac{\mu_0 I l_1}{2\pi} \ln \frac{a+l_2}{a}$

$$\therefore \varepsilon_i = -N \frac{d\Phi}{dt} = -N \frac{\mu_0 l_1}{2\pi} \ln \frac{a+l_2}{a} \cdot \frac{dI}{dt} = -N \frac{\mu_0 l_1}{2\pi} \ln \frac{a+l_2}{a} \times 10^3 \pi \cos(100\pi t)$$

代入 $t = 0.01$ 秒, 得: $\varepsilon_i = 8.7 \times 10^{-2} V$

6. $\varepsilon = Blv$ $\varepsilon = N(B_1lv - B_2lv) = Nlv\left(\frac{\mu_0 I}{2\pi a} - \frac{\mu_0 I}{2\pi(a+l)}\right) = 3 \times 10^{-3} V$